

Feature	Crossplane	Pulumi	Terraform
Declarative configuration	Yes (YAML)	Yes (Code in multiple languages)	Yes (HCL)
Self-healing control plane	Yes	No	No
Reconciling controllers	Yes	No	No
Extensibility	High (XRDs and Configuration Packages)	Moderate (Custom resources)	Moderate (Providers and modules)
Cloud provider support	Multi-cloud	Multi-cloud	Multi-cloud
Kubernetes integration	Yes	Partial	Partial
Programming languages	YAML	JavaScript, Python, Go, etc	HCL

A comparison of Crossplane, Pulumi, and Terraform

application, which is a sample NGINX deployment, to the Kubernetes cluster.

```

```yaml script example for deploying the web application with
nginx
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 replicas: 2
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:1.14.2
 ports:
 - containerPort: 80

apiVersion: v1
kind: Service
metadata:
 name: nginx-service
spec:
 selector:
 app: nginx
 ports:
 - protocol: TCP
 port: 80
 targetPort: 80
 type: LoadBalancer
```

```

This configuration in YAML will deploy your web application across three replicas and expose it through a LoadBalancer service.

Comparison of Crossplane, Pulumi, and Terraform

In recent years, Crossplane has gained popularity as a cloud-agnostic solution, despite the presence of competitors like Terraform and Pulumi. For a thorough understanding of Crossplane's capabilities, it is imperative to compare it to other prevalent IaC tools, including Terraform and Pulumi (refer to the Table above).

In summary, Crossplane is a robust container orchestration framework that provides declarative configuration, self-healing, extensibility, and controller reconciliation. It is an optimal option for developing cloud-agnostic and cloud-native solutions due to its seamless integration with Kubernetes and support for multi-cloud environments. Although Pulumi and Terraform also offer valuable IaC capabilities, Crossplane's distinctive assets are its comprehensive infrastructure management approach and its capacity to exploit the full potential of Kubernetes. **END** 🐼

References

- <https://www.crossplane.io/why-control-planes>
- <https://docs.crossplane.io/v1.19/guides/multi-tenant/>
- <https://docs.crossplane.io/v1.19/software/install/>
- <https://docs.crossplane.io/v1.19/getting-started/provider-azure/>
- <https://github.com/crossplane/crossplane/tree/main/design>

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